



ML IoT
HCMUT EE MACHINE LEARNING & IOT LAB

Qualcomm



DrowsyGuard

Edge-AI Driver Drowsiness Detection on Arduino UNO Q

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Platform: **Arduino UNO Q** – Qualcomm Dragonwing QRB2210 (Linux) + STM32U585 (MCU)

Qualcomm Future Makers: Hack The Challenge 2026 · Team ML_IoT_Love50

Agenda

⚠ PART 1

The Problem

What we are solving and who suffers from it.

📄 PART 2

The Product

What DrowsyGuard is, does and how it is built.

✅ PART 3

Technical Readiness

What already works and what is left.

👤 PART 4

Application Domains

Where it applies and how far it scales.



100% on-device

No internet needed

Privacy preserved

PART 1

The Problem We Solve



I Drowsy driving is a silent killer

⚠ Why it matters

- ▶ Fatigue and micro-sleep are among the leading causes of severe road accidents worldwide.
- ▶ Highest risk for **long-haul truck, coach and ride-hailing drivers**, especially at night.
- ▶ A micro-sleep of just **3-4 seconds** at 90 km/h means driving **blind for 100 m**.

🇻🇳 Vietnam context

Insert local statistics on fatigue-related traffic accidents here (source to be cited). The pain is concrete and nationwide.



A fatigued driver at the wheel, late at night.

Who suffers from it

Long-haul drivers

Night runs, tight delivery windows, hours alone at the wheel. The highest-risk group.

Coach & bus operators

Dozens of passengers per vehicle – one micro-sleep becomes a mass-casualty event.

Ride-hailing & private

Long shifts to hit income targets, often after a full working day.

And the businesses behind them

Fleet and logistics operators carry the cost: vehicle damage, cargo loss, insurance premiums, driver injury and reputational damage after every fatigue incident.

Existing solutions leave a gap

Approach	Limitation	Consequence
Built-in driver monitoring on premium cars	Only on expensive vehicles	Out of reach for most drivers
Cloud dashcam analytics	Needs network, adds latency	Fails on rural roads, feels intrusive
Wearables (smart bands, caps)	Uncomfortable, easy to forget	Low adoption in practice

🎯 The unmet need

A **low-cost add-on device** that runs **fully on-device**, works **offline**, protects **privacy**, and fits **any vehicle**.

PART 2

The Product

2

I DrowsyGuard in one line

💡 Concept

A small dashboard device that **watches the driver with a camera**, detects drowsiness **on the edge**, and triggers an **escalating alert** before it is too late.

🧠 Perceive

Retrained YOLO detects eyes, yawns and head nods in real time on the Linux core.

📈 Decide

PERCLOS, yawn rate and nod count are fused into a drowsiness level 0–3.

🔔 Act

The MCU escalates the alert: beep, then haptics, then full alarm and SOS.

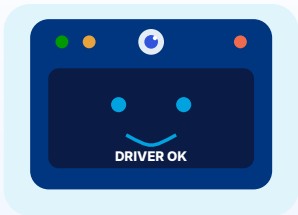
Low latency

Offline

Private by design

Fits any car

What the product looks like



Compact unit mounts on the dashboard, camera facing the driver.

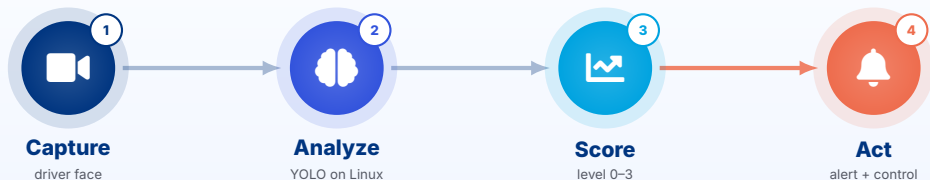
Simple status interface

- ▶ ● Green: driver alert.
- ▶ ● Amber: early drowsiness.
- ▶ ● Red: danger, full alarm.

Zero-friction UX

Plug into 12V power and it just runs. No pairing, no app setup, no driver action required.

How it works: end to end



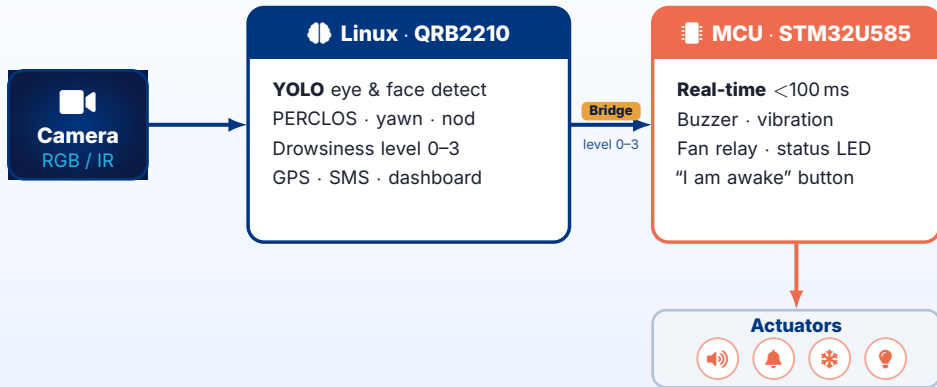
Everything stays on the device

No frame ever leaves the unit. All four steps run locally, so the system reacts instantly and keeps the driver's face private.

Reaction budget

Target end-to-end latency **under 200 ms** from eyes closing to the first alert – impossible to guarantee over the cloud.

System architecture: the dual-brain board



💡 Linux for heavy AI, MCU for instant control – one Arduino UNO Q board does both

Escalating alert ladder



Design principle

Gentle enough not to startle, strong enough to wake. The driver presses **"I am awake"** to reset.

Control functions

On the MCU – real time

- ▶ Buzzer and voice warning.
- ▶ Vibration motor in seat / steering.
- ▶ Relay for cool-air fan.
- ▶ Three-color status LED.
- ▶ “I am awake” acknowledge button.
- ▶ GPIO/CAN signal to cut throttle or flash hazards.

On Linux – connectivity

- ▶ Log drowsiness events with time + GPS.
- ▶ Send SOS via SMS / 4G if unresponsive.
- ▶ Fleet dashboard with per-driver safety score.
- ▶ Per-driver threshold personalization.

Fleet value

Logistics companies get a safety tool, not just a gadget.

Hardware and cost

Component	Role
Arduino UNO Q	Compute (AI + MCU)
RGB + IR camera	Day/night capture
Buzzer + speaker	Audio alert
Vibration motor	Haptic alert
Fan + relay	Cool-air wake-up
GPS + status LED	Location + UI



\$ Cost position

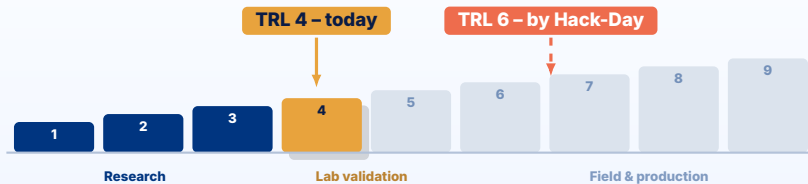
A **low-cost add-on**, a fraction of the price of built-in premium driver-monitoring systems. One self-contained unit, powered from the 12 V socket.

PART 3

Technical Readiness



Where we stand: Technology Readiness Level



✓ TRL 4 today – validated in lab

Detection pipeline proven on public datasets and on Arduino's own on-camera detector; alert logic specified end to end.

🚩 TRL 6 target – demo in a real cabin

With the loaned UNO Q we integrate camera, model and actuators into one unit and demonstrate it in a vehicle environment.

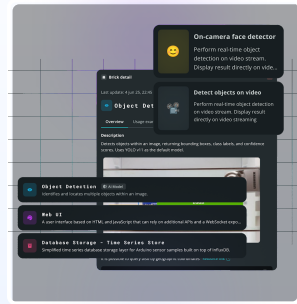
What already works today

✓ Proven building blocks

- ▶ **On-device detection on UNO Q** – proven by App Lab.
- ▶ **Datasets secured** – labelled, YOLO-ready.
- ▶ **Toolchain installed** – Arduino CLI, Flasher, AI Hub.
- ▶ **Alert logic specified** – PERCLOS + 4-level ladder.

✂ What the hardware unlocks

Measure real FPS, wire the actuators and close the loop in a cabin.



Arduino App Lab running an on-camera detector on the UNO Q itself.

AI model: a retrained YOLO detector

Detection classes

- ▶ **Eye open** vs **eye closed**
- ▶ **Yawn** (wide open mouth)
- ▶ **Head nod / down**

Why YOLO?

Fast single-shot detector – ideal for real-time inference on an edge device.

Day and night

RGB by day, infrared at night – works in a dark cabin.



The model tracks eye and mouth landmarks on every frame.

From frames to a drowsiness decision

PERCLOS = fraction of time the eyes are closed over a rolling 60-second window.



👁 PERCLOS

Rising closed-eye ratio is the strongest early sign of fatigue.

😊 Yawn rate

Frequent yawns add to the drowsiness score.

↓ Head nods

Repeated nodding pushes the level toward danger.

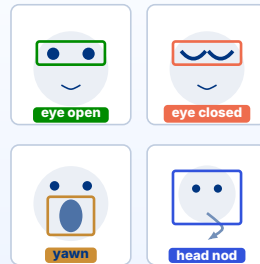
Training data

Dataset	Content	Use
Roboflow drowsiness / fatigue	Drowsy vs awake	Base classes
Roboflow eye state	Eyes open / closed	Class balance
Car-driver set	In-cabin footage	Fine-tuning

Data strategy

Merge public sets, **balance open vs closed**, augment for glasses, masks and low light. Exports directly to YOLOv8 and TFLite.

Annotated training frames



 YOLOv8-nano

Labelled eye/mouth frames teach the detector each state.

Optimized to run on the edge

⚙️ Model optimization

- ▶ YOLO **nano** backbone – small and fast.
- ▶ **INT8 quantization** for CPU/GPU inference.
- ▶ Runs on QRB2210 (no large NPU needed).

📊 State the numbers

Report a concrete figure, e.g. “*nano INT8 at $\approx N$ FPS on Cortex-A53*”. This proves it truly runs, not just in theory.



Tool: Qualcomm AI Hub for profiling

Risks and how we handle them

Risk	Mitigation
Dark cabin at night	Infrared camera + IR illumination
Driver wears glasses or a mask	Augment training data for both cases
Vibration, heat and dust in the cabin	Rugged mounting, automotive-grade parts
False alarms annoy the driver	Gradual ladder + "I am awake" acknowledge
Inference too slow on device	Nano backbone, INT8, resolution tuning

■ Roadmap – aligned to the competition



🚩 Our goal: a working UNO Q prototype demoed live at Hack-Day, 16 Aug 2026

PART 4

Application Domains

4

Primary domain: road safety

Commercial fleets first

- ▶ Logistics and long-haul trucking.
- ▶ Coach and inter-province bus operators.
- ▶ Ride-hailing and taxi fleets.

Why fleets are the entry point

They already pay for safety and telematics, buy in volume, and can measure the return directly in avoided incidents.



One device per cab, retrofitted to any vehicle.

■ The same technology travels far

Industry

Crane, forklift and heavy-machinery operators on long shifts.

Rail & marine

Train drivers and ship watchkeepers on monotonous night duty.

Mining & construction

Haul-truck drivers on remote sites with no network coverage.

Safety-critical roles

Control-room staff and security guards on night watch.

One core, many verticals

The same pipeline – **camera** → **on-device vision** → **fatigue score** → **escalating alert** – applies anywhere a human must stay alert. Only the mounting and the action at level 3 change.

Impact and how it scales

✓ Impact

- ▶ Fewer fatigue accidents – lives saved.
- ▶ Lower cost and insurance risk for carriers.
- ▶ Affordable safety for everyday vehicles.

Save lives

Cut costs

📈 Scale-up path

- ▶ Distraction alert (phone use at the wheel).
- ▶ Fleet-wide safety scoring.
- ▶ Insurance and telematics integration.

Ready to grow into a platform

I Demo scenario

Live demonstration

1. A volunteer sits in front of the device as "driver".
2. They slowly close their eyes / simulate a yawn.
3. L1: gentle beep and amber LED.
4. L2: loud alarm plus vibration and fan.
5. Press "I am awake" to reset to green.

✓ Proof plan

We show the escalation live on the board and log the event on the dashboard.

⚠ Backup

A short recorded demo video is prepared in case of hardware issues on stage.

🖼 Media placeholder

Drop a real prototype photo or screenshot here before the final round.

The team: ML_IoT_Love50



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AI & Computer Vision



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Vo Phuc Thinh

Connectivity & Cloud

Team synergy

End-to-end ownership: **camera** → **Edge AI** → **decision** → **MCU control** → **fleet dashboard**, built by one focused team.

In summary

Problem

Fatigue at the wheel kills, and affordable protection does not exist for ordinary vehicles.

Product

A dashboard device that sees drowsiness on-device and escalates the alert in under 200 ms.

Readiness

TRL 4 today, TRL 6 by Hack-Day: proven blocks, datasets and toolchain already in place.

Domains

Fleets first, then industry, rail, marine and mining – anywhere alertness is safety-critical.

What we ask for

The loaned **Arduino UNO Q**, Qualcomm SDK access and engineer mentorship – to turn a validated pipeline into a working in-cabin prototype by 16 August.

Thank You

Acknowledgements

We thank the Qualcomm Future Makers program and the organizers of Hack The Challenge 2026 for the opportunity, hardware and mentorship.